What Determines Household Size?

A Poisson GLM Analysis of Philippines FIES Data

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Background

- The Family Income and Expenditure Survey (FIES) provides nationally relevant information on household living conditions in the Philippines.
- Household size is an important demographic outcome because it is closely related to housing demand, food needs, and resource allocation.
- Understanding the factors associated with household size can help inform socio-economic analysis and public planning.

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Research Aim:

"To investigate which household-related variables influence the number of people living in a household"

Data Cleaning and Preprocessing

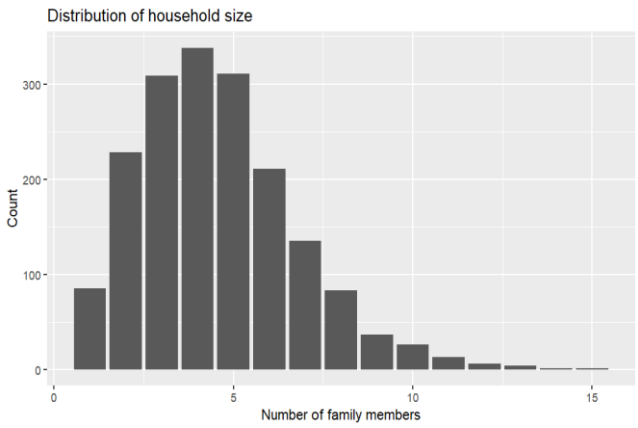
Featured Variables

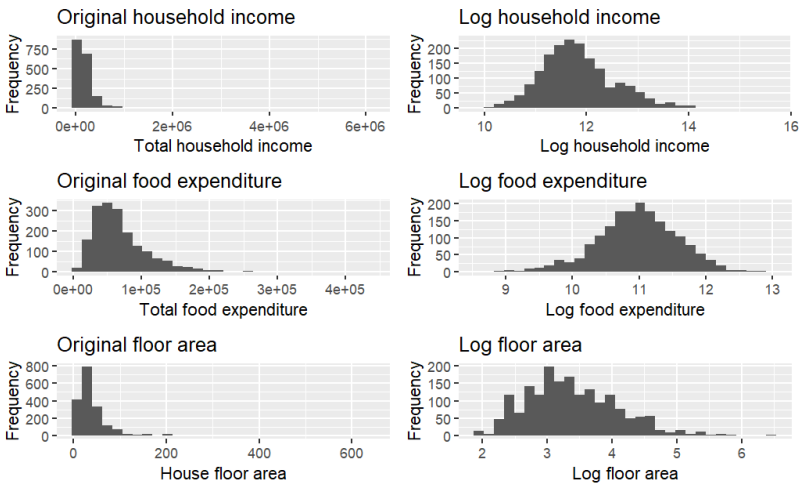
Variable	Type	Role
Total Family Members	Count	Response
Household Income	Numeric	Predictor
Food Expenditure	Numeric	Predictor
Head Age / Sex	Num/Cat	Predictor
House Type	Categorical	Predictor
Floor Area / Bedrooms	Numeric	Predictor
Electricity	Binary	Predictor

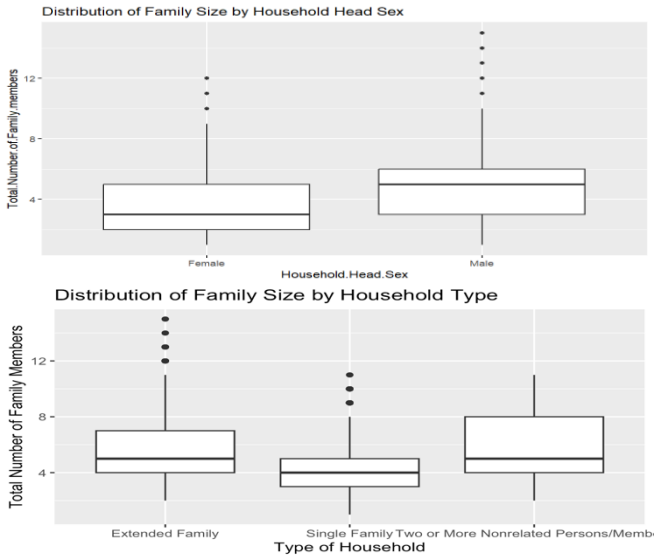
Preprocessing

- Data set has 11 Variables with 1788 observations; no missing values
- Categorical variables were retained as factors for model estimation.
- Log transformation applied to household income, food expenditure and floor area, reducing right-skewness.
- *Region was excluded because all observations came from the Zamboanga Peninsula, so it had no within-sample variation.*

Exploratory Data Analysis







Mean family size and variance, 4.55 and 4.79, respectively:

Suggests no strong overdispersion, supporting Poisson Regression.

After log transformation, the distributions of these variables become **more compact and regular**.

The transformed histograms show reduced skewness and a clearer concentration around central values.

Male-headed households tend to have larger household sizes than female-headed households.

Type of household indicated a difference in number of members.

GLM Modelling Strategy

Poisson Regression

Examine: **Household size and Socioeconomic characteristics**

Substantial Deviance Reduction

Food expenditure **strong positive** association with family size

Negative association of income with Household structure variables

Dwelling variables were **not statistically significant**

Overdispersion Validation

No evidence of Overdispersion: Poisson Model Appropriate

$$\frac{Residual\ deviance}{Residual\ df} = 0.585$$

Since the value is below 1

Negative Binomial

Estimated dispersion parameter: $(\boldsymbol{\theta} = \mathbf{120066.9})$

Large **θ** indicates that the negative binomial reduces to Poisson model

Negative Binomial AIC **slightly higher** than Poisson Model AIC

Chosen Model

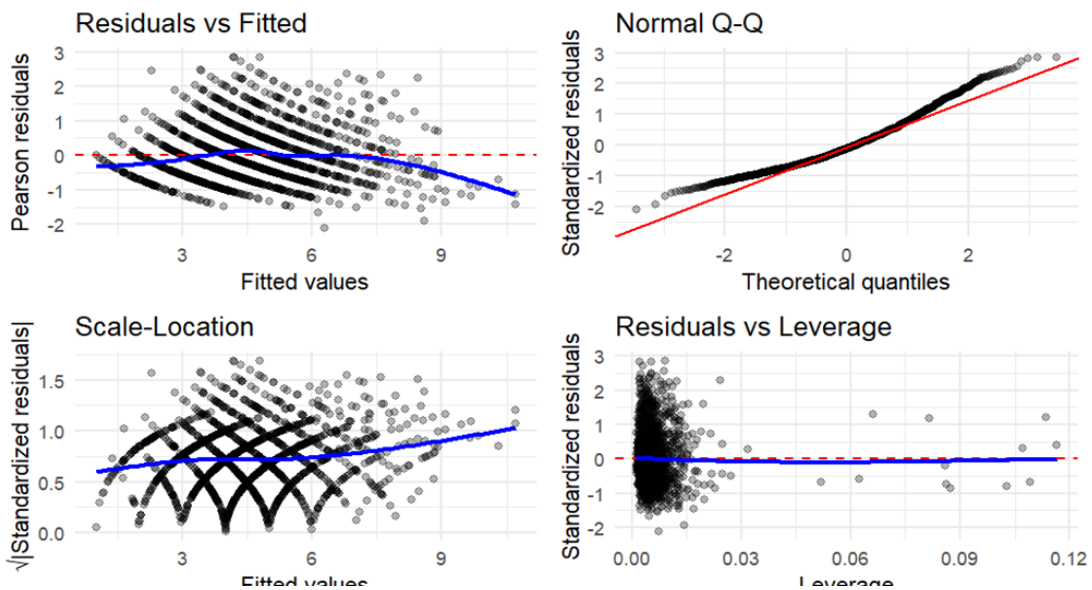
Poisson Specification:

$$Y_i | X_i \sim Poisson(\mu_i)$$

$$\log(\mu_i) = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_p X_{ip}$$

Diagnostics

Residual diagnostics



Residual plots indicate an acceptable fit overall, with mild deviations at higher fitted values but no severe influential points.

Multicollinearity

Term	VIF
log_income	5.01
log_food_exp	4.28
Number.of.bedrooms	1.69
log_floor_area	1.57
Household.Head.Age	1.29
ElectricityYes	1.25
Type.of.HouseholdSingle Family	1.21
House.Age	1.14
Household.Head.SexMale	1.12
Type.of.HouseholdTwo or More Nonrelated Persons/Members	1.02

VIF results suggest no serious multicollinearity, although log income and log food expenditure show moderate correlation.

Robustness

Sensitivity Check: Coefficients With and Without Food Expenditure

Variable	With Food	Without Food	Change	Sign Change
log_income	-0.380	0.162	0.542	Yes
Household.Head.Age	-0.002	-0.006	-0.004	No
log_floor_area	0.008	-0.012	-0.020	Yes

Purpose :

Test whether coefficients remain stable without food expenditure.

Conclusion:

The coefficient of log income changes substantially and reverses sign.

Result:

Most coefficients are stable, but the effect of income is sensitive to model specification.

Conclusions

Food expenditure, household type, income, and household head sex were the main associated factors.

Higher food expenditure was linked to **larger household size**, while single-family households tended to be smaller.

The **income effect was weak** and changed after excluding food expenditure, so it should be **interpreted with caution.**

Diagnostic checks and model comparison support the **Poisson model** as an appropriate baseline specification.

Thank you for your kind attention!

We would be happy to take any questions.